

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

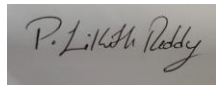
CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I P LIKITH REDDY (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Rubrics for Evaluation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks(100)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1.An image contains high-frequency noise.Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

ANSWER:

A Low-Pass Filter (LPF) is suitable for removing high-frequency noise.

Working:

The image is converted into the frequency domain using the Fourier Transform. The low-pass filter allows low-frequency components to pass while reducing or blocking high-frequency components. The filtered image is then converted back using the Inverse Fourier Transform.

Justification:

High-frequency components mainly represent rapid intensity changes, including noise. Therefore, suppressing these frequencies reduces noise.

Impact on image quality:

The image becomes smoother and less noisy, but excessive filtering may blur edges and fine details.

2.You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

ANSWER:

A Gaussian Low-Pass Filter is appropriate.

Working:

The Gaussian filter gives high importance to low-frequency components and gradually reduces high-frequency components. The frequency-domain image is multiplied by the Gaussian filter, followed by inverse Fourier transformation.

Justification:

Low-frequency components represent the general brightness and smooth structures of an image, while much unwanted noise occurs at higher frequencies. The filter preserves the important low-frequency information while reducing noise.

Result:

The image becomes smoother with reduced noise and fewer sharp artifacts.

3.An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.**Answer:**

A High-Pass Filter (HPF) is suitable for edge enhancement.

Working:

The image is transformed into the frequency domain. A high-pass filter suppresses low-frequency components and allows high-frequency components to pass. The inverse Fourier Transform produces the enhanced image.

Justification:

Edges and fine details correspond mainly to high-frequency components. Enhancing these frequencies makes boundaries more prominent.

Effect:

Edges become sharper and finer details are enhanced. However, excessive high-frequency enhancement can also amplify noise.

4.A smooth version of an image is required.Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.**Answer:**

A Low-Pass Filter, particularly a Gaussian Low-Pass Filter, can be used for image smoothing.

Working:

The Fourier Transform converts the image into frequency components. The low-pass filter retains low frequencies and attenuates high frequencies. The inverse Fourier Transform converts the result back into the spatial domain.

Justification:

High-frequency components represent sharp changes and fine details. Removing them reduces sudden intensity variations.

Result:

The image becomes smoother and less noisy, although edges and fine details may become blurred.

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Answer:

Thresholding-based segmentation is appropriate.

Working:

A threshold value is selected. Pixels with intensity greater than the threshold are classified as the object, while pixels below the threshold are classified as background.

$$g(x, y) = \begin{cases} 1, & f(x, y) > T \\ 0, & f(x, y) \leq T \end{cases}$$

Justification:

Since the object is significantly brighter than the background, intensity thresholding can easily separate them.

Effectiveness:

It is simple, fast, and effective when there is a clear difference in brightness.

6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Answer:

Otsu's thresholding method is suitable.

Working:

Otsu's method automatically determines an optimal threshold by analyzing the image histogram. It separates pixels into two classes: foreground and background.

Justification:

It is useful when the foreground and background have different intensity distributions. It avoids manually selecting a threshold.

Result:

A binary image is produced where one region represents the foreground and the other represents the background.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Answer:

The Canny Edge Detection method is suitable.

Working:

Canny edge detection involves:

1. Gaussian smoothing to reduce noise.
2. Gradient calculation to detect intensity changes.
3. Non-maximum suppression to obtain thin edges.
4. Double thresholding to identify strong and weak edges.
5. Edge tracking by hysteresis to produce connected edges.

Justification:

Canny provides accurate and thin edges while reducing the effect of noise.

Effectiveness:

It detects object boundaries clearly and generally produces better edge localization than simple gradient operators.

8.Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.**Answer:**

Region Growing segmentation is appropriate.

Working:

A seed pixel is selected. Neighboring pixels are examined and added to the region if their intensity values are sufficiently similar to the region's properties. The process continues until no suitable neighboring pixels remain.

Justification:

Region growing is specifically designed to group connected pixels with similar characteristics.

Result:

A meaningful region containing similar adjacent pixels is formed.

9.Detected edges in an image appear broken.

Explain and justify a method to improve edge continuity, including its role in post-processing.

Answer:

Morphological image processing, especially morphological closing, can improve edge continuity.

Working:

Morphological closing consists of dilation followed by erosion. Dilation connects nearby broken edge segments, while erosion restores the approximate original shape.

$$A \cdot B = (A \oplus B) \ominus B$$

Justification:

Small gaps and breaks between edge pixels can be filled using closing.

Role in post-processing:

It is applied after edge detection to connect fragmented boundaries and produce more continuous object outlines.

Result:

Broken edges become connected and object boundaries become easier to analyze.

10.You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Answer:

Boundary extraction using morphological operations is appropriate.

Working:

First, the object is segmented from the background. Then, morphological erosion is performed. The boundary can be obtained by subtracting the eroded image from the original binary image:

$$\text{Boundary}(A) = A - (A \ominus B)$$

where A is the binary object and B is the structuring element.

Justification:

Erosion removes the outer pixels of an object. Subtracting the eroded object from the original leaves only the boundary pixels.

Result:

A clear outline of the object is obtained, which can be used for shape analysis and object recognition.